We could keep going, but you should have gotten the feeling by now. The issue here is that we were focusing too much on details that have little importance. The key to get out of this impasse is to abstract out these details. This was obtained by defining a simplified computational model, the *RAM* model. The latter is a set of basic operations working on an internal *Random Access Memory*.

B.2 The RAM Model

Under the RAM model, a few assumptions hold:

- Each basic operation (arithmetic operations, `if`, function call) takes exactly a single one-time step (henceforth just referred to as *step*).
- Loops and subroutines are not considered simple operations. Instead, they are the composition of many one-time-step operations, and consume resources proportional to the number of times they run.
- Each memory access takes exactly one step.
- Memory is considered infinite.

The last point might seem a bit unrealistic, but notice that in this model we make no distinction between accesses to actual cache, RAM, hard drive or data-center storage. In other words

- For most real-world applications we can imagine to provide the memory needed
- With this assumption, we can abstract away the details of the memory implementation and focus on the logic.